

Workshop 02: Add MOSFET Thermal Behavior to the Simplified Buck Converter

Goal

Extend the simplified buck converter model with MOSFET thermal behavior, then use a second evaluation model to show the thermal effect more clearly. This workshop follows the clip

`02_Create_Temp_Simscape_Elec_Simplify.mp4`.

The clip has two sessions:

1. Start from `Create_Temp_Simscape_Elec_Simplify.slx` and finish with `Create_Temp_Simscape_Elec_Simplify_finished.slx`.
2. Open `Thermal_Mosfet_evaluation.slx` and use it to compare MOSFET temperature behavior when the heatsink convection setting changes.

By the end of this workshop, participants should be able to:

- Enable the thermal port on a Simscape Electrical MOSFET.
- Connect the MOSFET thermal port to a heatsink thermal network.
- Add a temperature sensor and route MOSFET temperature to the Scope.
- Run a coupled electro-thermal simulation.
- Use `Thermal_Mosfet_evaluation.slx` to make thermal differences easier to observe.

Prerequisites

- MATLAB with Simulink.
- Simscape.
- Simscape Electrical.
- Completed understanding of Workshop 01 buck converter operation.

Reference Values

Session 1: Thermal Buck Converter

Item	Value
DC input voltage	50 V
Duty cycle	0.8
PWM period	1e-5 s
Inductor	3e-3 H
Capacitor	1e-5 F
Load resistor	1 Ohm
MOSFET off conductance	1e-6 1/Ohm
MOSFET on resistance	0.01 Ohm
MOSFET thermal network	Junction-case
MOSFET thermal mass	[3, 25] J/K
Initial MOSFET junction temperature	25 degC
Heatsink convection area	1 m^2
Heat transfer coefficient	1 W/(m^2*K)
Heatsink mass	0.01 kg
Initial heatsink temperature	298.15 K
Ambient temperature	298.15 K
Simulation stop time	0.1 s

Session 2: Thermal Evaluation Model

Item	Value
DC input voltage	50 V
Load resistor	1 Ohm
Step time	1 s
Step initial value	0
Step final value	1
Heatsink convection area baseline	1 m ²
Heat transfer coefficient	1 W/(m ² *K)
Heatsink mass	0.01 kg
Ambient temperature	298.15 K
Simulation stop time	50 s

Session 1: Add Thermal Behavior to the Buck Converter

Start from:

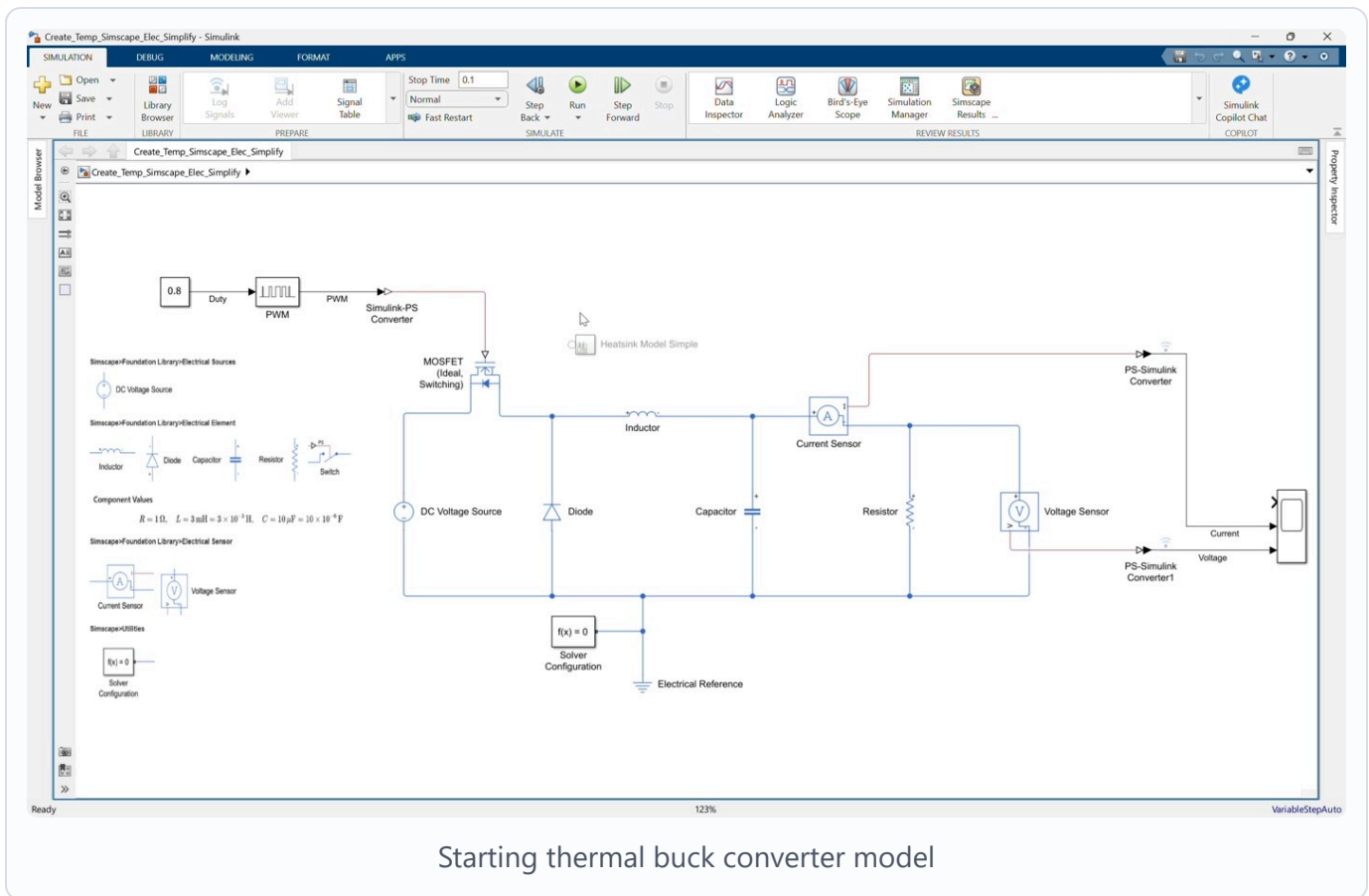
Create_Temp_Simscape_Elec_Simplify.slx

Finish with:

Create_Temp_Simscape_Elec_Simplify_finished.slx

The starting model already contains the electrical buck converter from Workshop 01. The Scope already has three inputs, but the first Scope input is not connected yet. In this session, that first Scope input is used for MOSFET temperature.

1. Open and review the starting model



Video screenshot: the starting simplified buck converter before adding the MOSFET thermal path.

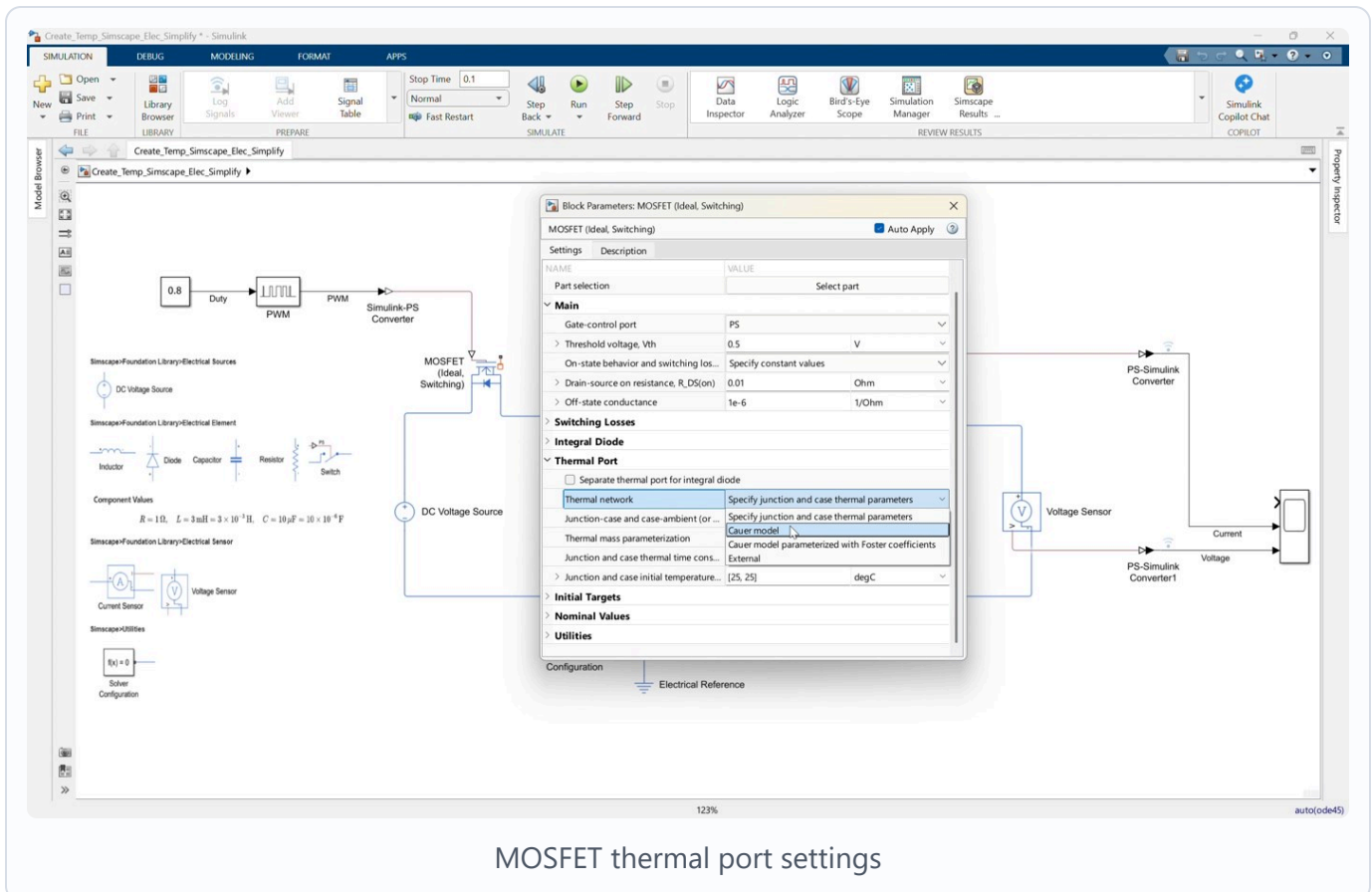
1. Open `Create_Temp_Simscape_Elec_Simplify.slx`.
2. Confirm the electrical buck converter is already complete:

- DC Voltage Source
- MOSFET (Ideal, Switching)
- Diode
- Inductor
- Capacitor
- Resistor
- Current Sensor
- Voltage Sensor
- PWM
- Simulink-PS Converter
- Two PS-Simulink Converter blocks
- Scope

- Electrical Reference
- Solver Configuration

1. Confirm the model still uses duty cycle 0.8.
2. Observe that the MOSFET does not yet expose a thermal conserving port.

2. Enable the MOSFET thermal port



*Video screenshot: enabling the MOSFET thermal network so the block exposes thermal port

H.*

1. Open the block parameters for MOSFET (Ideal, Switching).
2. Enable the thermal port or thermal network option.
3. Select the junction-case thermal network.
4. Confirm the key MOSFET thermal settings:

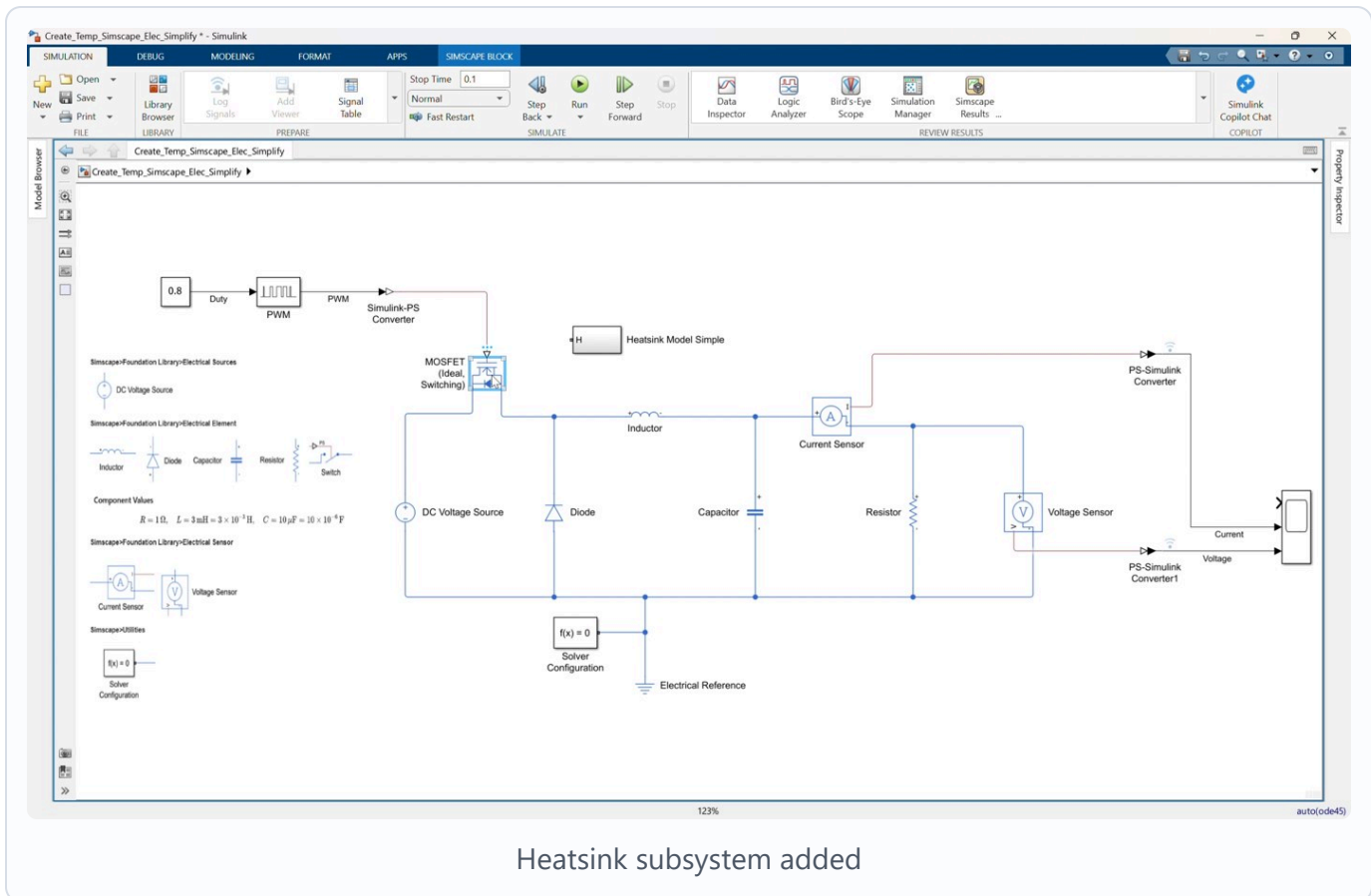
Thermal network = Junction-case

Thermal mass = [3, 25] J/K

Initial MOSFET junction temperature = 25 degC

1. Apply the parameter change.
2. Confirm the MOSFET block now shows a thermal conserving port **H**.

3. Add the heatsink subsystem

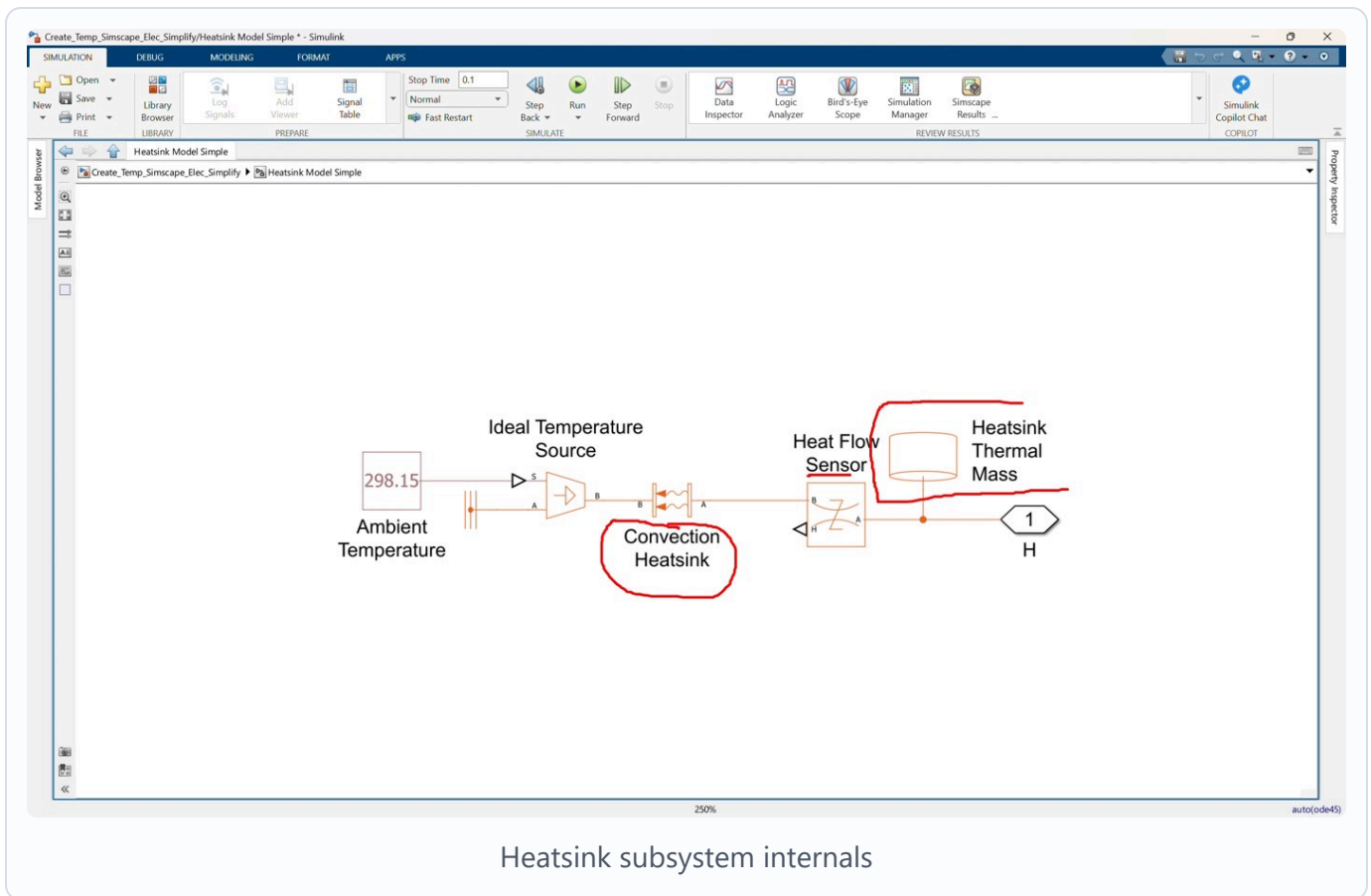


*Video screenshot: the **Heatsink Model Simple** subsystem used for MOSFET heat removal.*

1. Add or paste the subsystem **Heatsink Model Simple** into the model.
2. Confirm the subsystem has one thermal conserving port named **H**.
3. Connect the MOSFET thermal port **H** to the **Heatsink Model Simple** thermal port.

The heatsink subsystem represents heat removal from the MOSFET to ambient.

4. Review the heatsink subsystem



Video screenshot: internal thermal blocks inside the heatsink subsystem.

1. Open **Heatsink Model Simple**.

2. Confirm the subsystem contains:

- **Ambient Temperature**
- **Convection Heatsink**
- **Heat Flow Sensor**
- **Heatsink Thermal Mass**
- **Ideal Temperature Source**
- **Thermal Reference**

1. Confirm the main parameter values:

Ambient temperature = 298.15 K

Convection area = 1 m²

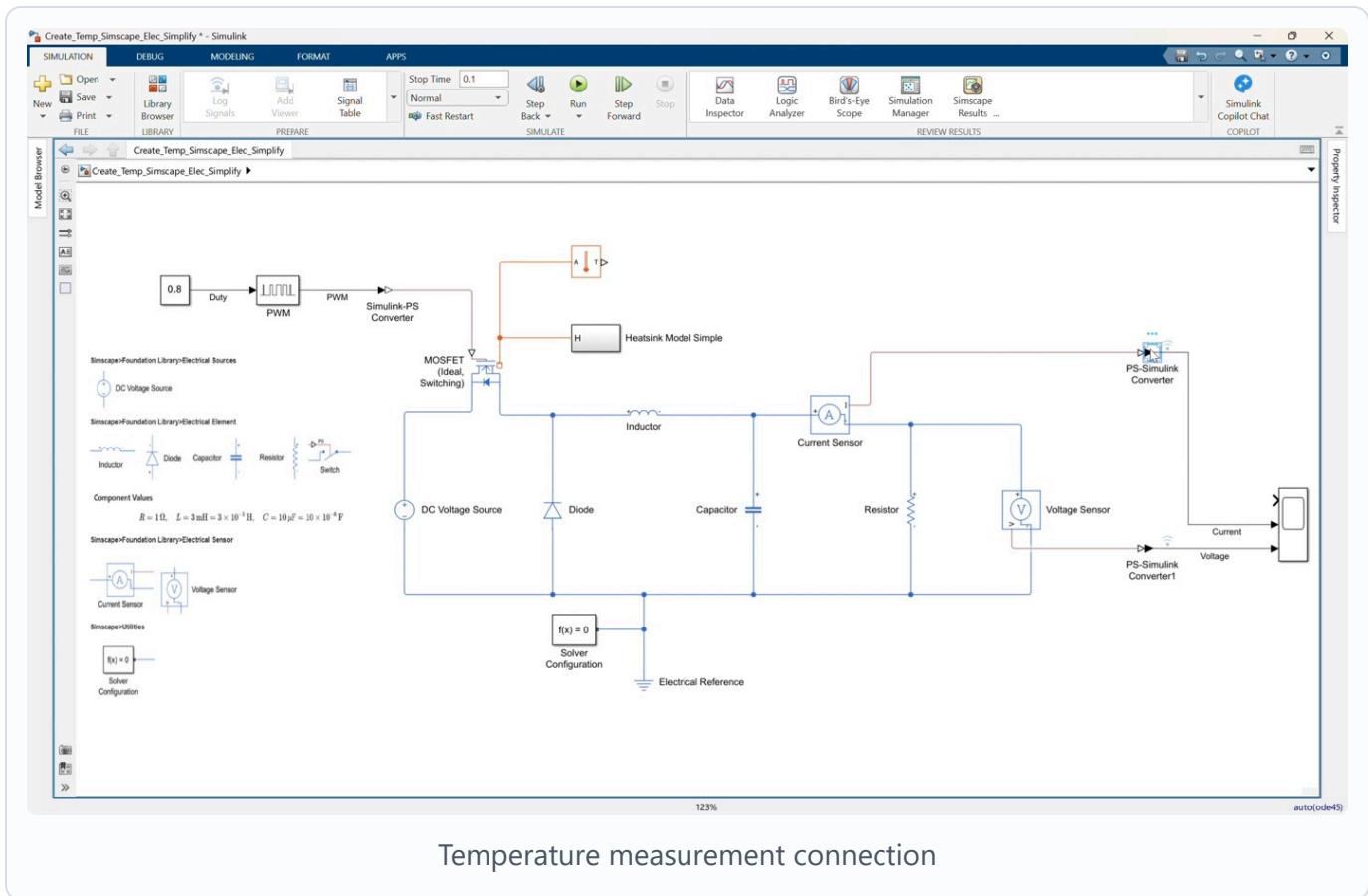
Heat transfer coefficient = 1 W/(m²*K)

Heatsink mass = 0.01 kg

Initial heatsink temperature = 298.15 K

1. Return to the top level of `Create_Temp_Simscape_Elec_Simplify.slx`.

5. Add MOSFET temperature measurement



Video screenshot: temperature measurement routed to the first Scope input.

1. Add a `Temperature Sensor`.
2. Connect the sensor thermal port to the same thermal node as the MOSFET thermal port and `Heatsink Model Simple`.
3. Add a `PS-Simulink Converter`.
4. Connect the temperature sensor physical signal output to the new converter.
5. Connect the new converter output to Scope input 1.

6. Name the signal **Temperature** .

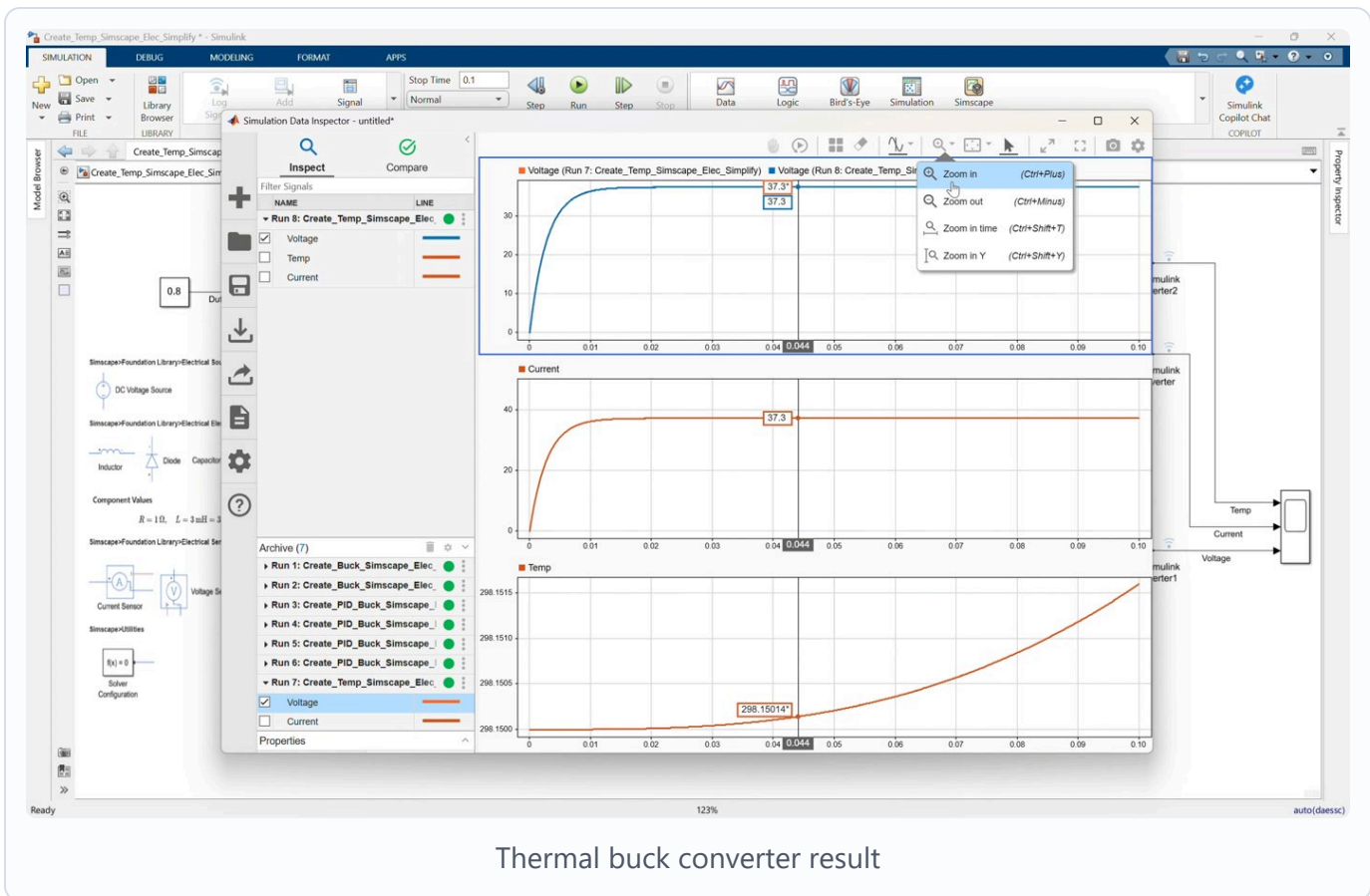
Suggested Scope signal order:

Scope input 1: Temperature

Scope input 2: Current

Scope input 3: Voltage

6. Configure and run the thermal buck converter



Video screenshot: simulation result after adding MOSFET temperature measurement.

1. Set the simulation stop time to:

0.1

1. Run the model.
2. Open the Scope.

3. Confirm the Scope shows:

- MOSFET temperature.
- Output current.
- Output voltage.

1. Save the completed model.

The result should match the structure of

```
Create_Temp_Simscape_Elec_Simplify_finished.slx
```

7. Explain the Session 1 result

Use this model to explain that the same electrical buck converter can be extended with a thermal domain. The MOSFET still works as the switching device, but its losses now create heat flow into the connected heatsink network.

The temperature change in this short `0.1 s` buck converter simulation may be small. The second session uses a simpler thermal evaluation model so the thermal effect is easier to see.

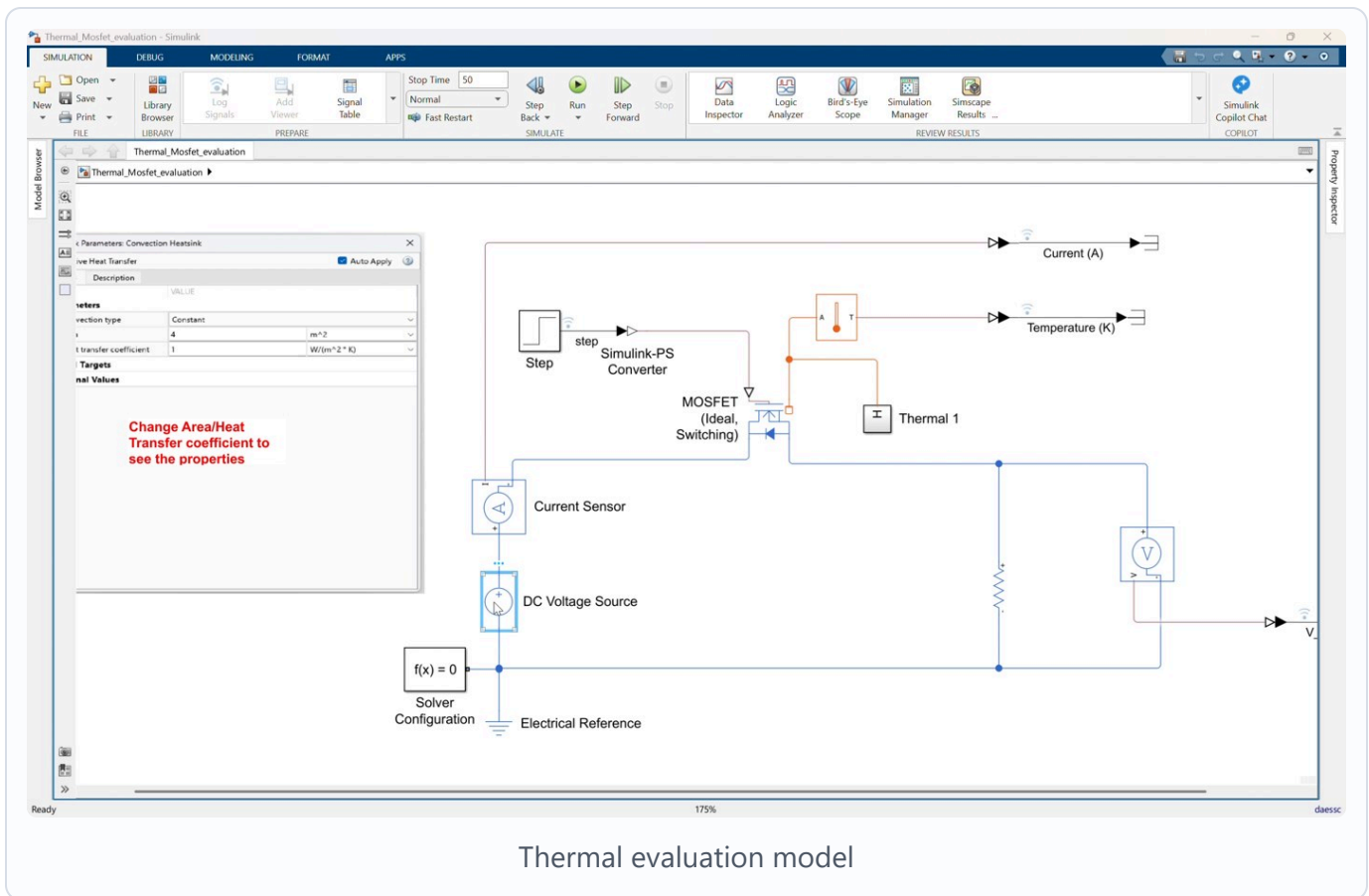
Session 2: Use the MOSFET Thermal Evaluation Model

Use this model:

```
Thermal_Mosfet_evaluation.slx
```

This model isolates the MOSFET, load, heatsink, and temperature measurement so convection changes produce a clearer thermal response over a longer simulation.

1. Open and review the evaluation model



Video screenshot: the simplified thermal MOSFET evaluation model.

1. Open Thermal_Mosfet_evaluation.slx .

2. Review the simplified circuit:

- DC Voltage Source : 50 V
- Current Sensor
- MOSFET (Ideal, Switching) with thermal port enabled
- Resistor : 1 Ohm
- Voltage Sensor
- Step gate command through a Simulink-PS Converter
- Temperature Sensor
- Thermal 1 heatsink subsystem

1. Confirm the Step block uses:

Step time = 1 s
Initial value = 0
Final value = 1

1. Confirm the simulation stop time is:

50

2. Inspect the Thermal 1 subsystem

1. Open the subsystem **Thermal 1**.
2. Locate the **Convection Heatsink** block.
3. Confirm the baseline settings:

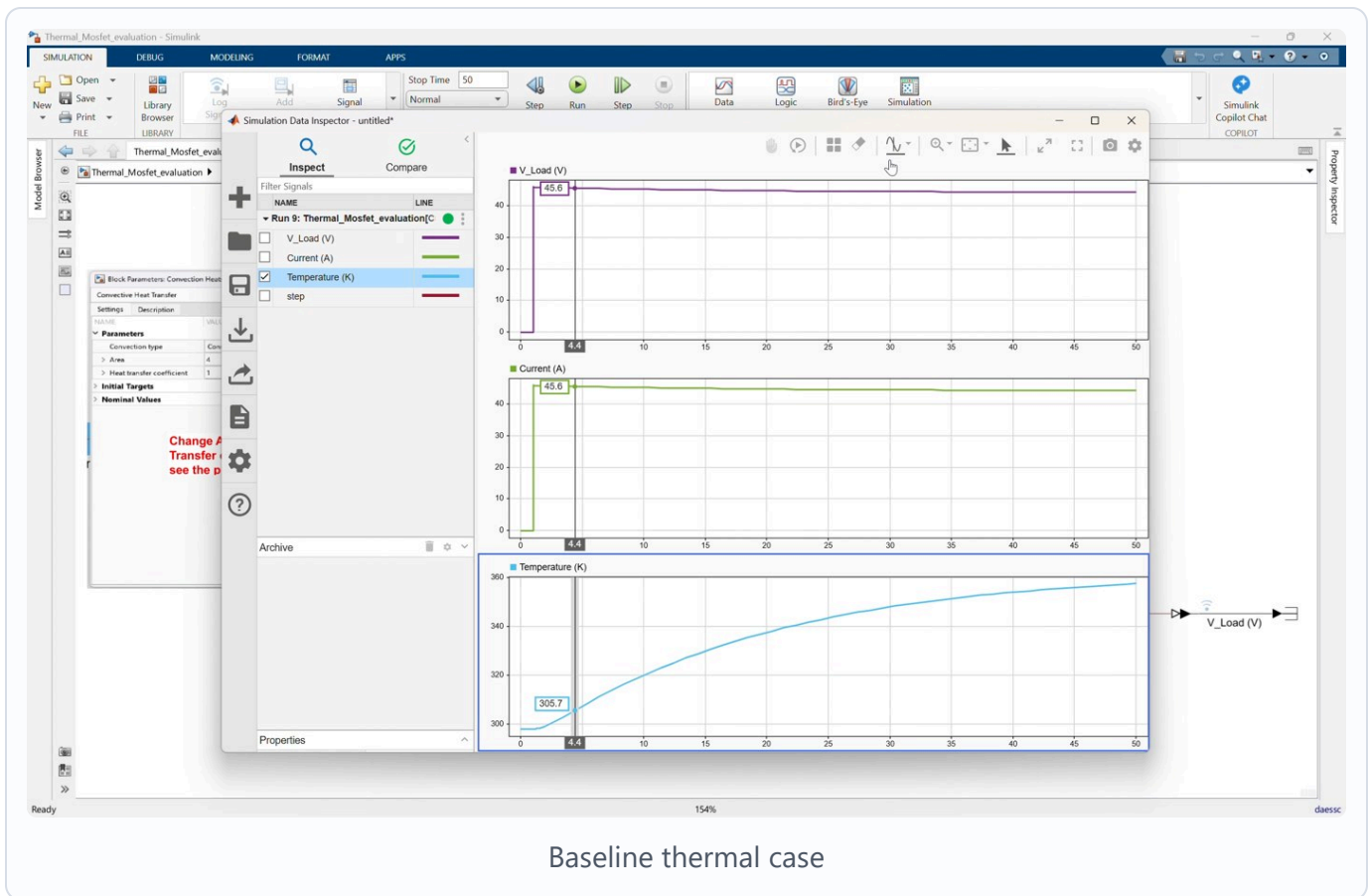
Convection area = 1 m²
Heat transfer coefficient = 1 W/(m²*K)

1. Confirm the heatsink thermal mass settings:

Heatsink mass = 0.01 kg
Initial heatsink temperature = 298.15 K

1. Return to the top level.

3. Run the baseline case

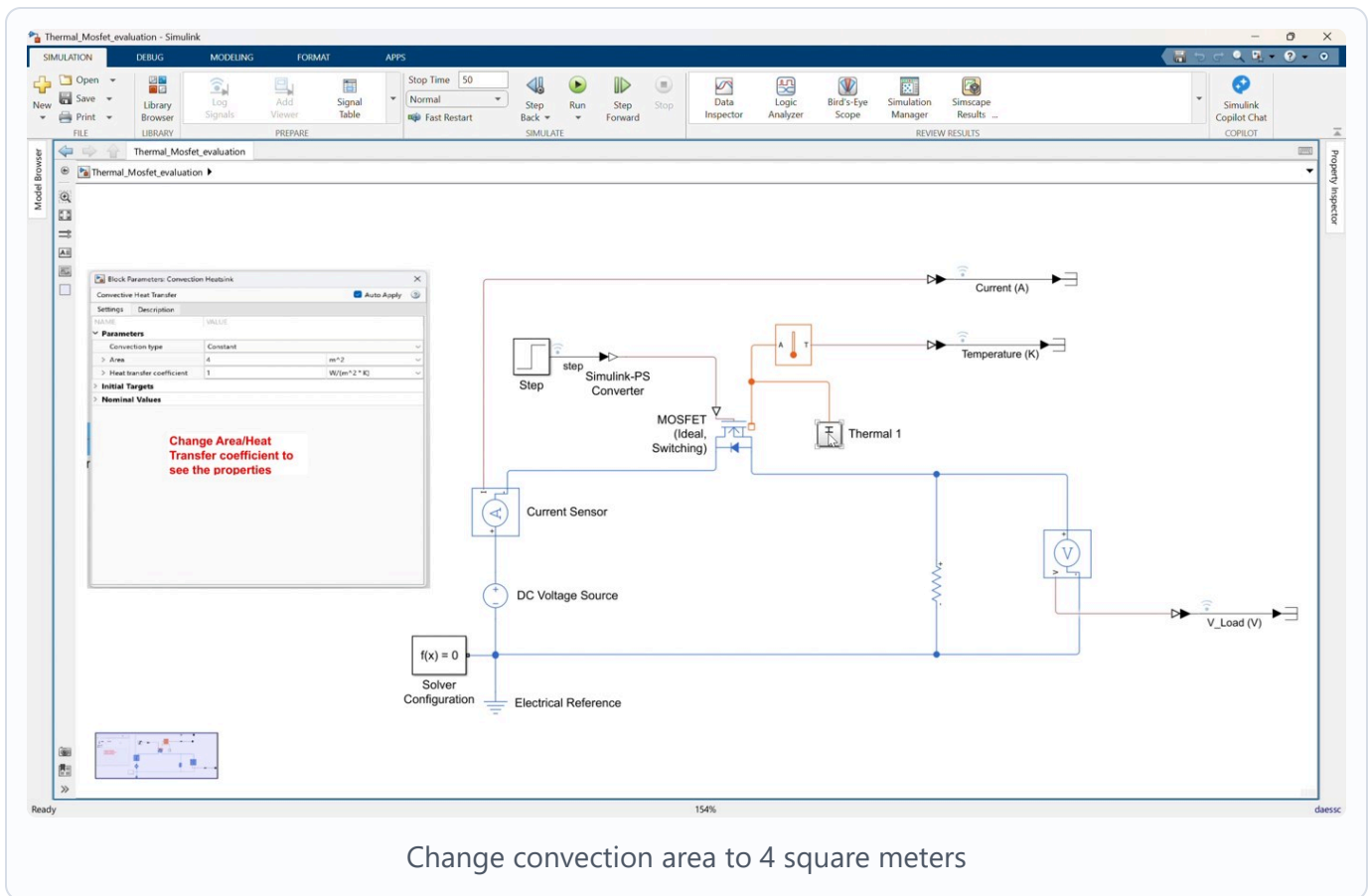


Video screenshot: baseline thermal evaluation run with convection area at 1 m^2 .

1. Keep the convection area at 1 m^2 .
2. Run the simulation.
3. Inspect the logged results for:
 - Temperature (K)
 - Current (A)
 - V_{Load} (V)

1. Use the baseline case as the reference cooling condition.

4. Increase the convection area



Video screenshot: changing the convection area for the improved cooling case.

1. Open **Thermal 1**.
2. Open **Convection Heatsink**.
3. Change the convection area from the baseline value to:

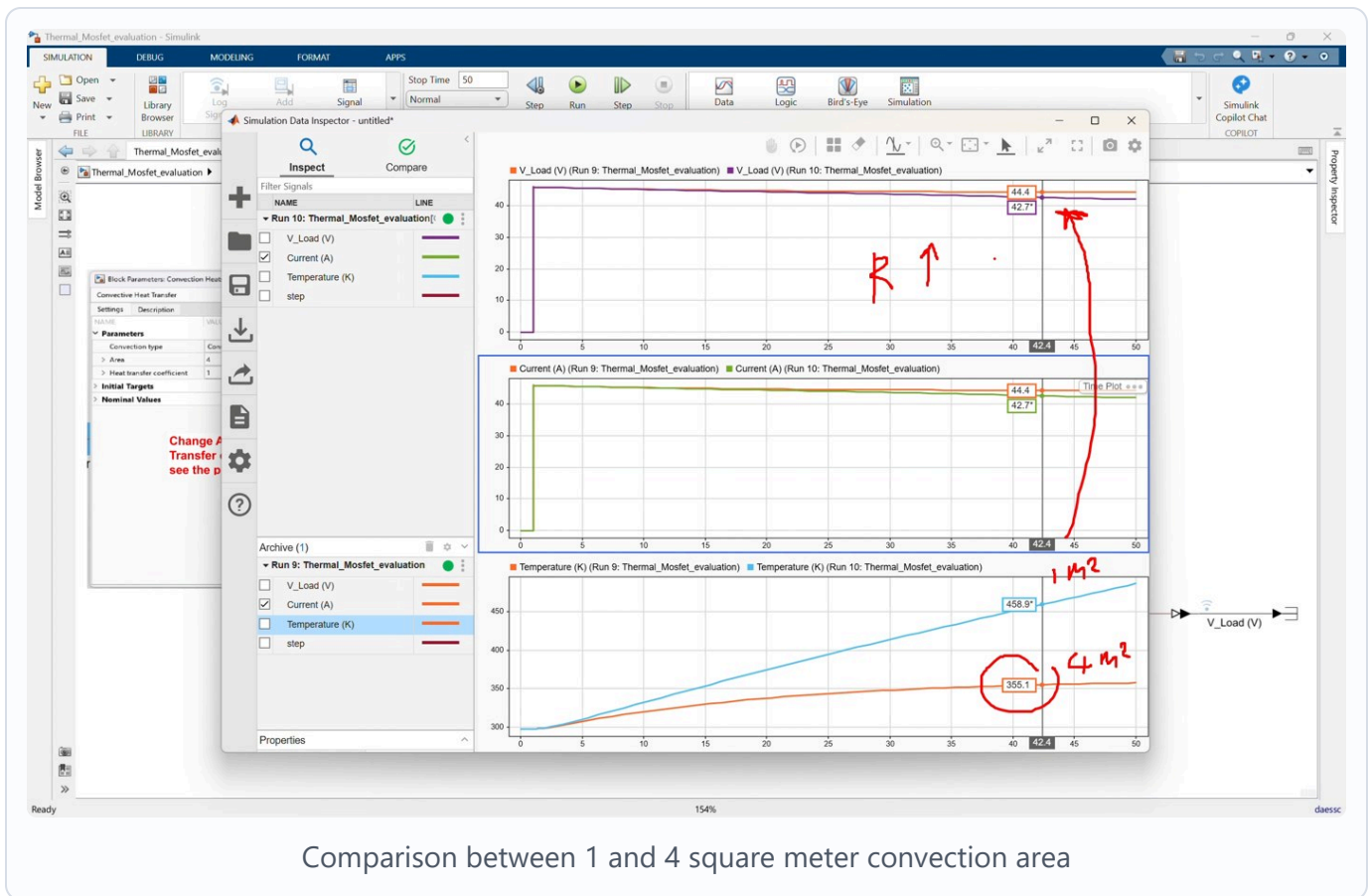
Convection area = 4 m²

1. Run the simulation again.
2. Compare the temperature response with the baseline case.

Expected observation:

- Larger convection area removes heat more effectively.
- MOSFET temperature rises less or settles at a lower value.
- The thermal effect is easier to see than in the short buck converter simulation.

5. Compare the two thermal cases



Video screenshot: comparison between the 1 m² and 4 m² convection-area runs.

Use the evaluation model to compare:

Case	Convection Area	Expected Temperature Behavior
Baseline cooling	1 m ²	Higher temperature rise
Improved cooling	4 m ²	Lower temperature rise

The exact temperature values depend on model settings and solver behavior. The important concept is the trend: stronger convection produces better cooling.

Suggested Instructor Flow

1. Open `Create_Temp_Simscape_Elec_Simplify.slx`.
2. Point out that the electrical buck converter is already complete.
3. Enable the MOSFET thermal port and explain the new thermal conserving port.
4. Add `Heatsink Model Simple` and connect it to the MOSFET thermal port.

5. Add the `Temperature Sensor` and route temperature to Scope input 1.
 6. Run the model and show that electrical and thermal behavior now simulate together.
 7. Open `Thermal_Mosfet_evaluation.slx`.
 8. Run the baseline case.
 9. Change the `Convection Heatsink` area and compare the temperature response.
 10. Close by explaining that thermal parameters such as convection area, heat transfer coefficient, heatsink mass, and ambient temperature can be changed directly in the physical model.
-

Participant Exercise

Ask participants to run two cases in `Thermal_Mosfet_evaluation.slx`:

Case	Convection Area
Baseline cooling	<code>1 m^2</code>
Improved cooling	<code>4 m^2</code>

For each case, record:

- Final MOSFET temperature.
- Peak MOSFET temperature.
- Whether the temperature rises quickly or slowly.

Then ask:

Which convection area gives the lowest MOSFET temperature?
Why does increasing convection area improve cooling?

Troubleshooting

Symptom	Likely Cause	Fix
MOSFET has no thermal port	Thermal port option is not enabled	Open MOSFET parameters and enable the thermal network/thermal port
Thermal network error	Heatsink subsystem is disconnected or missing thermal reference	Check <code>Heatsink Model Simple</code> or <code>Thermal 1</code> connections
Scope input 1 is empty	Temperature converter is not connected	Connect <code>Temperature Sensor</code> to <code>PS-Simulink Converter2</code> , then to Scope input 1
Temperature changes very little in Session 1	Buck converter simulation is short	Use <code>Thermal_Mosfet_evaluation.slx</code> for a clearer 50-second thermal comparison
Results do not clearly differ in Session 2	Convection area was not changed or the wrong run is being inspected	Compare the baseline <code>1 m^2</code> run with the updated <code>4 m^2</code> run

Reference Model Check

Session 1 completed model should include this added thermal path:

```

MOSFET thermal port H -> Heatsink Model Simple
MOSFET thermal node -> Temperature Sensor -> PS-Simulink Converter -> Scope
input 1
Existing Current Sensor -> PS-Simulink Converter -> Scope input 2
Existing Voltage Sensor -> PS-Simulink Converter -> Scope input 3

```

Session 2 evaluation model should support this comparison:

Step -> Simulink-PS Converter -> MOSFET gate

DC Voltage Source -> Current Sensor -> MOSFET -> Resistor -> Electrical Reference

MOSFET thermal port -> Thermal 1 subsystem

MOSFET thermal node -> Temperature Sensor -> PS-Simulink Converter

Change Thermal 1 / Convection Heatsink / area to compare cooling behavior